Reg. No. | | | | | | | | | | | [13AE751/12AE751]

SEVENTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, DECEMBER – 2016

COMPUTATIONAL FLUID DYNAMICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Draw neat sketches wherever required.
3.

1.	**a.**	Distinguish between conservative and non-conservative forms of conservation laws. Write the integral form of the continuity equation in non-conservative form.	**06**
	b.	Briefly indicate how 6 unknowns p, rho, u, v, w and e are evaluated for a perfect gas.	**04**
	c.	What are the advantages of integral and differential forms of conservation equations?	**04**
2.	**a.**	How are partial differential equations using characteristic method are classified and discuss about the computational aspects.	**06**
	b.	Classify:	**04**
		i. Euler equations of motion and	
		ii. Prandtl's boundary layer equations.	
	c.	Express $\frac{d^2u}{dx^2}$ and $\frac{d^2v}{dy^2}$ in central difference form.	**04**
3.	**a.**	Write briefly about any 3 methods for spatial discretization.	**06**
	b.	Discuss about some advantages of an unstructured grid generation.	**04**
	c.	How are boundary conditions important in the solution of any mathematical differential equation? Briefly write the boundary conditions for viscous flow past a flat plate.	**04**
4.	**a.**	What are the physical aspects to be kept in grid generation?	**04**
	b.	Briefly about adaptive grid generation.	**04**
	c.	Briefly discuss about O type, C type and H type grids.	**06**
5.	**a.**	What do you understand by 'pressure-correction' method of solution? How this is important for solution of incompressible flow?	**06**
	b.	What is 'aspect ratio' of a meshed element? Why should a large aspect ratio be avoided?	**02**

c. What is skewness of a mesh element? Why is it good to avoid a highly skewed element? **02**

d. Discuss briefly about numerical dissipation and numerical dispersion. **04**

6. a. Use the fourth order Runge-Kutta method to integrate $f(t,y) = 4e^{0.8t} - 0.5y$ using a step h=0.5 with y(0)= 2 from t=0 to 0.5. **10**

b. Discuss briefly about predictor-corrector methods for temporal discretization. **04**

7. a. What do you understand by Reynolds averaging? **04**

b. What are Reynolds stresses in turbulent flow? **04**

c. Discuss briefly about any two closure models for turbulence. **06**

8. a. Briefly write about normal shocks in the context of numerical computation. **04**

b. Using Thomas' algorithm, evaluate the unknowns T1, T2, T3 and T4 in the matrix given below. **10**

$$\begin{bmatrix} 3000 & -1000 & 0 & 0 \\ -1000 & 2000 & -1000 & 0 \\ 0 & -1000 & 1000 & -1000 \\ 0 & 0 & -1000 & 3000 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{Bmatrix} = \begin{bmatrix} 202500 \\ 2500 \\ 2500 \\ 802500 \end{bmatrix}$$